



Exposure assessment using human biomonitoring for glyphosate and fluroxypyr users in amenity horticulture



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ABSTRACT

Background: Pesticides and their potential adverse health effects are of great concern and there is a dearth of knowledge regarding occupational exposure to pesticides among amenity horticulturalists.

Objective: This study aims to measure occupational exposures to amenity horticulturalists using pesticides containing the active ingredients, glyphosate and fluroxypyr by urinary biomonitoring.

Methods: A total of 40 work tasks involving glyphosate and fluroxypyr were surveyed over the period of June – October 2015. Workers used a variety of pesticide application methods; manual knapsack sprayers, controlled droplet applicators, pressurised lance applicators and boom sprayers. Pesticide concentrations were measured in urine samples collected pre and post work tasks using liquid chromatography tandem mass spectrometry (LC–MS/MS). Differences in pesticide urinary concentrations pre and post work task, and across applications methods were analysed using paired *t*-tests and linear regression.

Results: Pesticide urinary concentrations were higher than those reported for environmental exposures and comparable to those reported in some agricultural studies. Log-transformed pesticide concentrations in post-work samples had a geometric mean (geometric standard deviation) of 0.66 (1.11) $\mu\text{g L}^{-1}$ for glyphosate and 0.29 (1.69) $\mu\text{g L}^{-1}$ for fluroxypyr. Linear regression revealed a statistically significant positive association to exist between the time-interval between samples and the log-transformed adjusted (i.e. post- minus pre-task) pesticide urinary concentrations ($\beta = 0.0039$; $p < 0.0001$).

Conclusion: Amenity horticulturalists can be exposed to pesticides during tasks involving these products. Further research is required to evaluate routes of exposure among this occupational group.

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1. Introduction

The use of pesticides and their possible health effects is of significant concern, especially with an estimated annual usage of 12.5 million tonnes of pesticides worldwide (Michael and Alavanja, 2009). Occupational exposures to pesticides have regularly been associated with adverse health effects such as cancer, respiratory diseases, detrimental reproductive health, and neurological diseases (Acquavella et al., 2003; Arbuckle et al., 2001; Bretveld et al., 2008; Guyton et al., 2014; Henneberger et al., 2014; Hernandez and

Menendez, 2016; Lauria et al., 2006; Montano, 2014; Mostafalou and Abdollahi, 2013, 2016; Weidner et al., 1998; Ye et al., 2013).

Pesticide products must be approved before use and, dependent on their approval, they can be used in a variety of settings. However, occupational exposure assessment studies have largely been confined to agriculture for example, farm workers (Baharuddin et al., 2011; Blanco et al., 2005; Krenz et al., 2015; Lebailly et al., 2009; MacFarlane et al., 2008; Rubino et al., 2012; Singleton et al., 2015; Strong et al., 2008) and their families or residents near farmland (Curwin et al., 2007; Galea et al., 2015; Hanchenlaksh et al., 2011; Sams et al., 2016).

There is a comparably less research on occupational exposures to pesticides in horticulture; in greenhouses (Bouvier et al., 2006; Flores et al., 2011; Machera et al., 2003; Machera et al., 2002;

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Nuyttens et al., 2009; Ramos et al., 2010), florists (Bouvier et al., 2006) and amenity horticultural workers (Johnson et al., 2005). This sector contributes approximately €90 billion to the European economy (EPSO, 2014). The horticulture industry is diverse and can be divided into commercial (production of food and flowers for commercial gain) and amenity (for recreational or ornamental purposes) horticulture (The Chartered Institute of Horticulture, 2016).

Pesticide products are used to control weeds and scrub growth in amenity areas and for the preservation and growth of decorative plants or endangered species in ornamental horticulture (Illing, 1997). However, the use of pesticides in horticulture is different to agriculture, so results from agricultural based studies are not always applicable to the horticultural sector. Horticulturists regularly use a variety of pesticides for a range of applications from protecting indoor and outdoor plants from disease or insect infestations, to control nuisance and invasive weed growth in public areas. Horticultural workers tend to use a great variety of pesticides in a wider variety of applications throughout the year.

These differences in industrial use of pesticides can include the dose rate of pesticide products used, the type of work tasks and the application frequency. Use of pesticides is not always confined to the growing season and horticultural workers usually use a variety of application methods including boom sprayers, manual knapsack sprayers as well as battery operated sprayers on a more regular basis. In horticulture, a work task involving pesticide use typically involves: the collection of the product from the chemical store, mixing of the product to the correct dose rate, loading the applicator (mixing and loading), spraying the pesticide and finally cleaning the application equipment after use (MacFarlane et al., 2013).

Previous studies on horticultural workers spraying herbicides in residential and public areas, or on spraying pesticides in greenhouses suggest that dermal exposure is the prominent exposure route, accounting for up to 99% of the total body exposure (Aprea et al., 2004; Flack et al., 2008; Tuomainen et al., 2002; Vitali et al., 2009). Potential dermal exposure (PDE) levels tend to vary depending on the stage of the activity (i.e. mixing & loading, spraying or cleaning stage), application method used, condition of the applicator, human factors e.g. worker's attitude towards safety (Johnson et al., 2005; Ramos et al., 2010) and use of personal protective equipment (PPE) (Lander and Hinke, 1992; MacFarlane et al., 2013; Nigg et al., 1986). Higher exposures have been associated with the mixing and loading activity (Ramos et al., 2010) especially when more than one mix and load task is completed in a day (Flores et al., 2011). Higher exposures have also been associated with the use of high pressure application methods (Machera et al., 2003) or when using a hand held lance walking forward into the pesticide mist (Nuyttens et al., 2009) and from exposure due to surface transfer of previously applied pesticide contaminants (MacFarlane et al., 2013). Although PDE data provides useful information on exposure, PDE estimates do not always reflect the total exposure risk from other routes, such as, inhalation and inadvertent ingestion.

Biological monitoring, measuring pesticides or their biomarkers in samples such as blood or urine, can be used to evaluate total uptake of a pesticide by an operator by all exposure routes and can provide a more reliable estimate of pesticide exposure where information on pharmacokinetics is available (Acquavella et al., 2003; Chester, 2010). Biological monitoring has previously been used in studies evaluating occupational exposures to pesticides in both the agricultural (Acquavella et al., 2004; Aprea, 2012; Hanchenlaksh et al., 2011; Mesnage et al., 2012; Rubino et al., 2012; Singleton et al., 2015) and horticultural sectors (Aprea et al., 2005; Baldi et al., 2006; Bouvier et al., 2006; Flores et al., 2011). Recent papers have called for more exposure assessment studies, as exposure can vary greatly between workers in apparently similar conditions. This information would provide direct evidence of exposure levels to evaluate

the range of human pesticide exposure, to advance epidemiology and provide more reliable data for predicting health effects, to provide regulatory estimates of exposure and to ensure safe and sustainable pesticide use (Acquavella et al., 2003; Gangemi et al., 2016).

Fluroxypyr and glyphosate are commonly used pesticides in amenity horticulture, fluroxypyr is an herbicide used to control broadleaf weeds and woody brush, glyphosate is a non-selective weed killer. Fluroxypyr is classified as a 'not likely' human carcinogen (US EPA, 1998) and has been assigned an Acceptable Daily Intake (ADI) and an Acceptable Operator Exposure Level (AOEL) of 0.8 mg/kg body weight per day (bw/day) (EFSA, 2011). Glyphosate, is the highest volume herbicide used worldwide, contained in over 750 pesticides (Guyton et al., 2014) and was classified as 'Group 2A – probably carcinogenic to humans' (IARC, 2015). The European Food Safety Authority conducted a peer review and concluded that it was 'unlikely to pose a carcinogenic hazard to humans' (EFSA, 2015). Glyphosate has an ADI of 0.5 mg/kg bw/day and an AOEL of 0.1 mg/kg bw/day (EFSA, 2015).

A small number of studies have used biological monitoring methods to evaluate human exposures to glyphosate (Acquavella et al., 2004; Conrad et al., 2017; Curwin et al., 2007; Jayasumana et al., 2015; Krüger et al., 2014; Mesnage et al., 2012) but none among amenity horticulture workers. In addition, to the author's knowledge, thus far, there have been no human biomonitoring studies on fluroxypyr.

The aim of the current study was therefore to characterise the occupational exposures via a newly developed biomonitoring method for both glyphosate and fluroxypyr in amenity horticultural workers.

2. Materials and methods

2.1. Site description and study population

This study was conducted over the period of June – October 2015 in the Republic of Ireland, at field sites managed by Commissioners for Public Works in Ireland – referred to hereafter as the Office of Public Works (OPW). The OPW manage and maintain a diverse portfolio, which includes national monuments and heritage services ranging from ornamental gardens to the Phoenix Park located in Dublin which is 707 ha and one of the largest walled city parks in Europe (OPW, 2010).

A walk through survey was performed by the researcher at the selected OPW sites including national parks, ornamental gardens and historic monuments, to collect information on the frequency of use of pesticides containing glyphosate and fluroxypyr and spraying methods used. Four similar exposure groups (SEGs) were defined based on the pesticide and application method used (Table 1). Biological monitoring study protocols were developed following a review of previous research protocols (Galea et al., 2011; Health and Safety Authority, 2011; Health and Safety Executive (HSE) UK, 1997). Study protocols were approved by the National University of Ireland, Galway Research Ethics Committee (Ref 15/May/04).

The lead researcher presented the project aims and objectives and distributed project information leaflets at different sites and events hosted by the OPW with the aim of informing potential study participants. Following these events, amenity horticulturists that used relevant pesticides and worked with the OPW at the designated sites, were invited to participate in the study. Recruitment was done in coordination with the OPW Health and Safety Unit, however participation was voluntary. Prior to the commencement of the study, the lead researcher met with the workers and presented an overview of the sampling protocols and meth-

Table 1
Characteristics of established similar exposure group (SEG).

Similar Exposure Group	Application Method	Pesticide product (Concentration of A.I.)	Volume/Concentration of pesticide used in application
<i>Glyphosate with a manual knapsack.</i>	A sprayer with a handheld lance connected to a container that is carried on the back and dispenses pesticide spray from a manual lever.	Roundup Biactive XL (360 g/L) Pistol (250 g/L) Destrol Amenity (250 g/L) Glyphos (360 g/L) Rambo 360 (360 g/L)	1 part concentrate to 10 parts water, used for spot spraying, spraying footpaths and chemical edging of gardens.
<i>Glyphosate with a controlled droplet applicator.</i>	Controlled droplet application produces spray droplets that are relatively uniform in size and permits the applicator to control droplet size. This can assist in reducing drift.	Nomix Dual (120 g/L)	This is a premixed solution. This applicator is used for footpaths and clearing areas for total weed control.
<i>Glyphosate with a pressurised handheld lance (between 2 and 6 bar pressures).</i>	A modification of the boom sprayer to use a hand held lance at 2–3 bar pressure.	Roundup Biactive XL (360 g/L)	The pressurised and motorised lance would use 1 part concentrate to 10 parts water and is used for total weed control of large open areas and courtyards.
	A motorised Hardi portable applicator with a handheld lance at 2–3 bar pressure.	Pistol (250 g/L)	The portable SteelMaster V5 is used to spray weeds above head height such as spraying the walls of a historic buildings to maintain the integrity of the structure.
	51 SteelMaster V5 professional sprayer, which is a pressurised manual handheld lance at 6 bar pressure.	Glyphos (360 g/L) Rambo 360 (360 g/L)	
<i>Fluroxypyr using a boom spraying attached to a tractor.</i>	Tractor mounted 8 or 16 m boom sprayer at 2–3 bar pressure.	Synero (100 g/L)	1 part concentrate to 10 parts water for large parkland areas.
		Praxys (100 g/L)	

ods. Informed consenting workers completed a self-administered questionnaire to collect contextual information on the participants including their work activities, out-of-work use of pesticides, dietary habits and their smoking status.

3. Biological monitoring

3.1. Urine collection

A convenience sampling approach was followed for the collection of relevant urine samples. Surveys were completed on days that workers were using glyphosate or fluroxypyr based pesticides as part of their normal work duties. The researcher attended the sites on each of the application days to observe the activities and obtain relevant contextual information to inform the urinary concentration results.

On the day workers participated in the study, they were asked to provide two urine samples (up to 50 mL). The first was a background sample taken before the work task began (pre samples) and the second was provided within one hour of the entire task completion (post sample). To prevent possible contamination during sample collection, participants were provided with written instructions on how to provide a urine sample and they were asked to wash their hands before giving a sample. Once the participant had provided a sample, the sample was tightly sealed and placed in a clean sealed plastic bag. The researcher stored all samples in a dedicated portable freezer in the field, and used disposables gloves when handling all samples. Urine samples were labelled with a unique identifier number, date, time and the active ingredient involved and stored at -18°C in a dedicated project freezer until laboratory analysis. The sampling time in this study was defined as the time interval between the pre and post sample collection. An activity diary was completed for each task by the researcher through direct observations. This included information on the sampling time, application method, pesticide used, personal protective equipment (PPE) used and climatic conditions. Any changes in PPE equipment used by the workers, increases in wind speed or

change in weather conditions as well as breaks taken and problems encountered during the tasks by the workers were also recorded.

3.2. Urine sample analysis

All samples were transferred and analysed by the Health and Safety Laboratory (HSL), Buxton, UK. An analytical method was developed during this project for both glyphosate and fluroxypyr, which was adapted from a method used for food and beverage extraction of glyphosate (Nagatomi et al., 2013). The method presented here is based on the analysis of the unchanged parent compounds of glyphosate and fluroxypyr in urine. Analytes were extracted from urine by solid phase extraction (SPE) with subsequent analysis by liquid chromatography tandem mass spectrometry (LC–MS/MS). Changes to the original extraction method (Nagatomi et al., 2013) include the substitution of hydrochloric acid with formic acid in the elution stage of SPE with no requirement to filter samples before analysis. This is the first reported method for determining fluroxypyr in human urine.

3.3. Standards and reagents

Standards of glyphosate, glyphosate- $2\text{-}^{13}\text{C}$ ^{15}N , and fluroxypyr were purchased from Sigma Aldrich (Dorset, UK). 2,4,5-trichlorophenoxyacetic acid- d_4 (2,4,5-T- d_4) was purchased from qm_x (Essex, UK). HPLC grade methanol and acetonitrile were purchased from Rathburn Chemicals (Walkerburn, UK). HPLC grade acetic acid was purchased from Fisher Scientific (Loughborough, UK) and formic acid was purchased from Sigma-Aldrich. Water was purified by an Elga water filtration system (Veolia, High Wycombe, UK).

3.4. Sample preparation

For the analysis of glyphosate and fluroxypyr, 500 μL and 250 μL of sample urine was diluted with 500 μL and 750 μL of deionised water, respectively. The prepared samples were spiked with 10 μL of internal standard (500 $\mu\text{g/L}$), glyphosate- $2\text{-}^{13}\text{C}$, ^{15}N

for glyphosate and 2,4,5-T-d₄ for fluroxypyr. SPE was performed on Strata SAX (1 mL/100 mg) cartridges (Phenomenex, Macclesfield, UK) which were initially washed with 1 mL of methanol, follow by conditioning with 1 mL deionised water. The 1 mL of prepared sample was loaded onto the cartridge and subsequently washed with 1 mL deionised water and 1 mL methanol. The cartridges were dried and the analytes eluted into polypropylene tubes with 2 × 500 µL aliquots of 10% formic acid in methanol. Samples were dried under a stream of nitrogen with heat blocks set at approximately 40 °C. Samples were then reconstituted in 100 µL of 1% acetic acid before analysis by LC–MS/MS.

3.5. Instrument analysis

Analysis was performed on an Applied Biosystems API3200 mass spectrometer (ABSciex, Warrington, UK) coupled to a Shimadzu HPLC system (Milton Keynes, UK). Chromatography was carried out on a Zorbax SB-C3 150 × 4.6 mm, 5 µm (Agilent, Stockport, UK) column equipped with a C18 guard column (Phenomenex). Mobile phases used were 1% acetic acid (mobile phase A) and acetonitrile (mobile phase B). Twenty microliters of the sample was injected into the system and separation achieved using a gradient elution at a flow rate of 0.4 mL/min. The gradient programme was as follows: 0 min (B: 5%), 2 min (B: 5%), 8 min (B: 95%), 10 min (B: 95%), 10.5 (B: 5%), 16 min (B: 5%). A 6 port valve directed the flow to waste and then to the ion source at 2 min.

The mass spectrometer source temperature was set at 500 °C and ion spray voltage at –4500 v. Curtain gas, collision gas, ion source gas 1 and gas 2 flow rates were set at 45, 4, 40 and 40 L/min respectively. Mass spectrometry analysis was performed in negative MRM mode. Mass transitions and compound specific parameters are given in Supplementary Information, Table 1.

The method was linear over the calibration range 0–20 µg L^{–1} ($R^2 > 0.999$) for both analytes. Intra and inter assay coefficient of variation (CVs) of 3.54% ($n = 10$) and 9.96% ($n = 40$, over 4 runs) were achieved for glyphosate and 2.81% ($n = 10$) and 9.99% ($n = 30$) for fluroxypyr, respectively. The limits of detection (LOD) were set at 0.5 µg L^{–1} and 0.1 µg L^{–1} for glyphosate and fluroxypyr respectively based on observed peaks with a signal to noise ratio of $\geq 3:1$. Creatinine analysis was also completed on all urine samples using an automated alkaline picrate method using a Pentra 400 clinical analyzer (Cocker et al., 2011).

Statistical analysis for this study was performed using STATA software (Stata 14.1, 2016). Prior to the analysis, all exposure measurements with levels below the LOD were imputed, in R software (R Software, 2016). A single random imputation method was applied to substitute left censored measurements with a value between 0 and the LOD randomly generated from the log normal distribution (Helsel DR., 2005). Urinary pesticide concentrations (both simple and corrected for creatinine values) approximated a log normal distribution and thereby all analysis was performed on naturally log-transformed data. Consequently, distribution characteristics are summarised by geometric means (GM) and geometric standard deviations (GSD) along with arithmetic mean (AM), minimum and maximum levels. Statistical analysis was conducted using biological monitoring data expressed as pesticide urinary concentrations and repeated with the creatinine corrected values. For the latter analysis, samples with creatinine concentrations below 2 mmol L^{–1} or greater than 30 mmol L^{–1} were excluded from the statistical analysis (Cocker et al., 2011; EWDTS, 2002), which resulted in the exclusion of four samples from the statistical analysis. To determine whether there was a significant difference between the log-transformed pre and post task samples, a paired students *t*-test was conducted. Analysis of Variance (ANOVA) was used to examine differences in urinary pesticide concentrations between the four SEGs, whereas relationships between

Table 2

Demographics, work and sampling characteristics. Data are presented as number of observations (percent) or mean values (range) for parameters on a continuous scale.

Variable		
Participant and work characteristics	Number (N)	Percentage (%)
Participants	18	100
Male	17	94
Female	1	6
Paired Samples (Pre & Post)	40	100
...Workers that sprayed pesticides the day before the task	23	58
Participants that smoke, n	1	6
Age (years)	Mean	Range
	44.5	33–66
Measurement characteristics	Mean	Range
Length of time working with pesticides (years)	19.7	0–40
On average, months of the year the worker uses pesticides	5.2	0–7
Length of the task time in each SEG (minutes)		
Spraying glyphosate with a manual knapsack	40	5–115
Spraying glyphosate with a large droplet applicator	81	33–195
Spraying glyphosate with a pressurised applicator	133	24–375
Spraying fluroxypyr using a boom sprayer	149	75–364
...Sampling time per SEG (minutes) (Pre to post sample)		
Spraying glyphosate with a manual knapsack	110	45–378
Spraying glyphosate with a large droplet applicator	235	45–445
Spraying glyphosate with a pressurised applicator	219	80–445
Spraying fluroxypyr using a boom sprayer	199	110–375
Participants pesticide & PPE use	Participant Number (N)	Percentage (%)
Glove use – Yes	17	94
Tyvek use – Yes	12	67
RPE use – Yes	15	83
Which tasks did the workers complete at least once?		
Collection of the pesticide from the store	13	72
Mixing and loading the chemical	9	50
Spraying the chemical	16	89
Cleaning the equipment	6	33
...Did the worker take a break during the task?		
Yes	5	28
Did the worker reuse personal protective equipment (PPE)?		
Reuse PPE	16	89
Do not reuse PPE	3	17
No Answer	1	6
Uses pesticides outside of work	15	83
Similar exposure groups (SEGs)	No. of participants	No. of samples
Spraying glyphosate with a manual knapsack	8	10
Spraying glyphosate with a large droplet applicator	4	7
Spraying glyphosate with a pressurised applicator	5	14
Spraying fluroxypyr using a boom sprayer	5	9

work characteristics, tasks and urinary pesticide concentrations were examined in classical linear regression. When appropriate, Mann-Whitney tests were also applied to examine differences in continuous variables.

Table 3
Biological monitoring results ($\mu\text{g L}^{-1}$) describing concentrations of glyphosate and fluroxypyr in urinary samples collected pre and post to the working activities as well as adjusted concentrations for baseline levels (i.e. post minus pre concentrations). Results are presented overall and per similar exposure group.¹

Variable	k	n	<LOD N (%)	AM	GM	SD	GSD	Min	Max
Combined glyphosate SEGs									
Adjusted	17	31	14 (45)	0.64	0.28	1.45	2.12	−0.32	7.23
(Post–Pre samples)									
Pre sample	17	31	20 (65)	0.71	0.42	0.92	0.94	0.13	3.43
Post sample	17	31	14 (45)	1.35	0.66	2.18	1.11	0.12	10.66
Manual knapsack (glyphosate)									
Adjusted	8	10	5 (50)	0.29	0.15	0.66	1.67	−0.09	2.13
(Post–Pre samples)									
Pre sample	8	10	5 (50)	0.83	0.51	0.93	1.03	0.13	3.2
Post sample	8	10	5 (50)	1.12	0.62	1.55	1.09	0.17	5.33
Pressurised lance (glyphosate)									
Adjusted	5	14	6 (43)	0.41	0.31	0.73	1.98	−0.32	2.60
(Post–Pre samples)									
Pre sample	5	14	11 (79)	0.53	0.34	0.79	0.79	0.17	3.2
Post sample	5	14	6 (43)	0.94	0.57	1.13	1.00	0.12	4.09
Controlled droplet applicator (glyphosate)									
Adjusted	4	7	3 (43)	1.62	0.56	2.71	2.92	−0.04	7.23
(Post–Pre samples)									
Pre sample	4	7	4 (57)	0.90	0.48	1.20	1.16	0.15	3.43
Post sample	4	7	3 (43)	2.53	1.00	3.90	1.40	0.28	10.66
Tractor mounted boom sprayer (fluroxypyr)									
Adjusted	5	9	3 (33.3)	0.73	0.36	1.08	2.23	−0.04	2.63
(Post–Pre samples)									
Pre sample	5	9	3 (33.3)	0.11	0.10	0.05	0.49	0.05	0.19
Post sample	5	9	3 (33.3)	0.84	0.29	1.09	1.69	0.04	2.74

*k = number of participants in the group.

n = number of samples in the group.

AM = Arithmetic Mean, GM = Geometric Mean, SD = Standard deviation, GSD = Geometric standard deviation, Min = Minimum, Max = Maximum.

4. Results

4.1. Demographic and working characteristics

The study population consisted of 18 horticulture amenity gardeners, who applied pesticides as part of their work. All the participants had the minimum legally required training on the safe use of pesticides under Irish legislation (Department of Agriculture Food & the Marine, 2016). Demographic characteristics of participants in this study are presented in Table 2. The participants were predominantly male (94%). Only one participant stated on their questionnaire that they currently smoked tobacco products.

Spraying with a manual knapsack was performed by eight workers, whereas four workers sprayed pesticides using a large droplet applicator, five using a pressurised applicator, and another five using a boom sprayer, some workers participated in more than one similar exposure group. In about half of the total tasks sampled (58%), a pesticide task had also been completed the previous day. Of the 40 tasks analysed in this study, nearly all workers conducted the spraying sub-task (93%) with fewer completing the mixing and loading sub-task (53%). This is mostly because the solution was premixed for the controlled droplet applicator, thus, no mixing and loading was required for those tasks. The sampling time for the tasks completed in this study ranged from 45 min to 375 min. The vast majority of these workers wore PPE for each task, such as gloves (94%), respiratory protective equipment (RPE) (83%) and over half wore Tyvek suits. Most of the workers (89%) also reused their PPE. The majority of workers did not take breaks during the pesticide task (78%).

4.2. Levels of urinary pesticide concentrations

A summary of the biomonitoring data for glyphosate and fluroxypyr, expressed as pre and post samples urinary concentration ($\mu\text{g L}^{-1}$), and adjusted urinary pesticide concentrations (post sample minus pre sample concentrations), are presented in Table 3.

The biological monitoring results expressed as creatinine corrected pesticide urinary concentrations in $\mu\text{mol/mol}$ creatinine are summarised in Table 2 of the online Supplementary information. A total of 80 samples were collected in this study, 40 pre-exposure and 40 post-exposure samples.

Of the pre samples collected, 58% (23 samples) were below the LOD and of the post samples 43% (17 samples) were below LOD. The levels of urinary excretion across SEGs are shown in Fig. 1.

4.3. Differences in exposure across SEGs, sampling and working parameters

There was a statistically significant difference between the log transformed pre and post sample (a paired *t*-test, $p < 0.001$); for both $\mu\text{g L}^{-1}$ and $\mu\text{mol/mol}$ creatinine, urinary pesticide concentrations. No statistical difference in log-transformed pre- and post-sample concentrations were observed between the four SEGs means ($p = 0.38$), with only 9% of the variance attributed to differences between SEGs. Average concentrations were similar between SEGs but variability in the levels of exposure within individual SEGs was large (Fig. 1).

When the effects of sampling time, defined as time between collection of the pre and post urinary sample, on the log-transformed pesticide urinary concentration (post-pre) were examined in linear regression, a positive statistically significant association was observed ($\beta = 0.0039$; $p < 0.0001$). Similar results were found with log-transformed creatinine corrected values ($\beta = 0.0030$; $p < 0.0001$). The model explained almost half ($r^2 = 0.45$; $p < 0.0001$) of the variation in pesticide urinary concentrations (Fig. 2) and

¹ The 90th percentiles of the post exposure values in microgram per litre ($\mu\text{g L}^{-1}$), of the groups with 10 or more samples were as follows: The 90th percentiles for the glyphosate group is $4.09 \mu\text{g L}^{-1}$ ($n = 31$). The individual glyphosate groups 90th percentile are $3.35 \mu\text{g L}^{-1}$ ($n = 10$) for the manual knapsack group and $2.80 \mu\text{g L}^{-1}$ ($n = 14$) for the pressurised lance group.

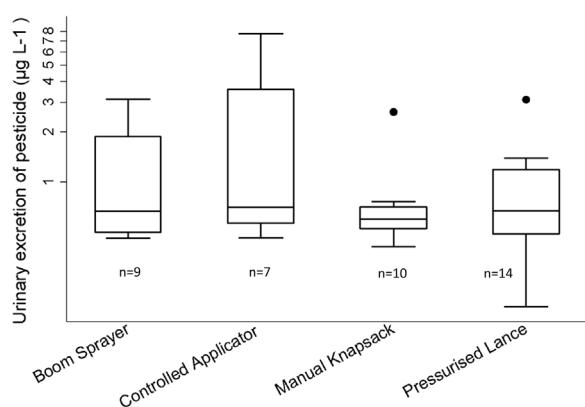


Fig. 1. Boxplot showing the adjusted pesticide ($\mu\text{g L}^{-1}$) concentrations (i.e. post exposure concentration minus the pre exposure concentrations) in micrograms per litre in each SEG. The box is the 25th to the 75th percentile, the line within the box is the median and the whiskers the lower and the upper adjacent values. Single points indicate outliers.

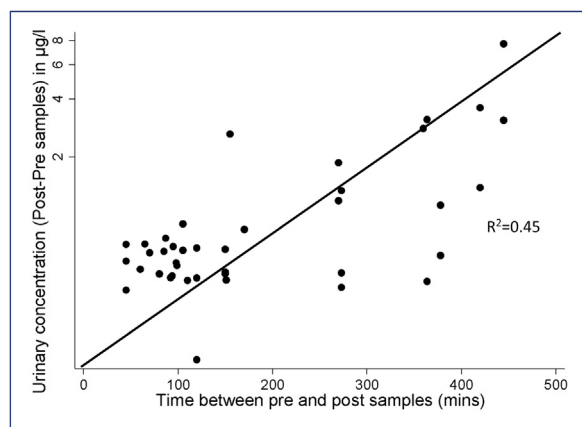


Fig. 2. Time period between pre and post samples and the urinary concentration of pesticides.

slightly less with the creatinine corrected samples ($r^2=0.37$; $p<0.0001$). When results were stratified by SEG, the positive effects of sampling time on log-transformed urinary excursions were strongest ($\beta=0.0075$; $p<0.0001$) for measurements in the boom sprayer group, which was the only group using fluroxypyr. Similar but somewhat less pronounced associations were observed also for the controlled droplet applicator ($\beta=0.0053$; $p<0.05$) and pressurised lance ($\beta=0.0029$; $p<0.05$) groups, but not for measurements performed during spraying with manual knapsack ($\beta=0.0011$; $p=0.5$).

Interestingly, log-transformed pesticide urinary concentrations (post-pre) in measurements from workers who took breaks during the task were significantly higher ($\beta=0.54$; $p<0.05$), by a factor 1.7 [i.e. $\exp(\beta)$] on average, compared to those from workers who did not take breaks. Trends were similar when analysis was repeated using the creatinine corrected values ($\beta=0.47$; $p<0.05$). Similarly, median sampling times were significantly longer for the measurements where workers took breaks compared to those who did not (273 versus 105 mins) (Mann-Whitney test, $p<0.01$), as well as those samples that had pesticide concentrations above the LOD compared to those with concentrations below the LOD (170 versus 98 mins), (Mann-Whitney test, $p<0.01$).

5. Discussion

This study provides human biomonitoring data for workers employed as amenity horticulturists spraying glyphosate and fluroxypyr. As previously discussed, studies quantifying occupational exposure to pesticides in amenity horticulture are very sparse. There is limited published human biomonitoring data for glyphosate (Acquavella et al., 2004; Conrad et al., 2017; Curwin et al., 2007; Jayasumana et al., 2015; Krüger et al., 2014; Mesnage et al., 2012; Niemann et al., 2015), (Table 4), and to the authors knowledge, there is no published human exposure data for fluroxypyr.

A straightforward, reproducible and robust method, capable of determining both glyphosate and fluroxypyr was developed based on a previous method available by Nagatomi et al. (2013). Sensitivity of the assay ($0.5 \mu\text{g L}^{-1}$ for glyphosate, $0.1 \mu\text{g L}^{-1}$ for fluroxypyr) was comparable to those previously reported for glyphosate $0.1 \mu\text{g L}^{-1}$ (Conrad et al., 2017) to $7.5 \mu\text{g L}^{-1}$ (Honeycutt and Rowlands, 2014).

The study results suggest that amenity workers have elevated urinary pesticide levels for both glyphosate and fluroxypyr, above what would be expected from dietary exposures. Of the post exposure samples, pesticide concentrations in 43% (17 samples) were less than the LOD. Almost all of the post exposure samples pesticide concentrations were higher than concentrations in the pre task samples, however all post exposure levels were low. Although no human biomonitoring data are available for the Irish population, arithmetic mean exposure values (post task urine sample) for glyphosate reported in this study of $1.35 \mu\text{g L}^{-1}$ are higher than mean values of $0.21 \mu\text{g L}^{-1}$ reported in an European environmental exposure study (Hoppe, 2013). Similarly, both mean and maximum values reported in our current study ($1.35 \mu\text{g L}^{-1}$ and $10.66 \mu\text{g L}^{-1}$, respectively) are also higher compared to the maximum urinary glyphosate concentrations ($0.41 \mu\text{g L}^{-1}$) reported by Conrad et al. (2017) for German adults with non-occupational exposures to glyphosate.

Compared to previous studies among working populations, our reported glyphosate exposure concentrations are comparable to those reported by Mesnage et al. (2012) and Curwin et al. (2007) on agricultural exposures. Mesnage et al. (2012) reported urine glyphosate concentrations 0.95 to $4.35 \mu\text{g L}^{-1}$ with a peak value of $9.5 \mu\text{g L}^{-1}$ for a farmer who sprayed glyphosate from a tractor and using a hand held spraying. Likewise our reported levels are slightly lower but within the range of exposure values reported by Curwin et al. (2007) and Acquavella et al. (2004) in studies among farm families in Iowa (range of measured concentrations 0.02 – $18 \mu\text{g L}^{-1}$) and farmers in Minnesota and South Carolina, (GM $1.4 \mu\text{g L}^{-1}$ and GM $7.9 \mu\text{g L}^{-1}$, respectively).

The mean urinary pesticide concentrations were comparable across all four SEGs studied. However, exposure data from those workers of the controlled droplet applicator group appeared to be somewhat more variable than those from the other groups. Workers viewed the controlled droplet applicator as a safer application method as it dispenses large droplets and reduces the risk of spray drift. In addition, it does not require mixing and loading of the concentrate prior to application (Nomix Enviro, 2016). The observed exposure variation within this SEG therefore may reflect inconsistent use of PPE and complacent work practices (e.g. adjusting the nozzle during the task), due to the workers perception that this applicator is low risk. Protective coveralls were only worn for

² Liquid chromatography tandem-mass spectrometry.

³ High-performance liquid chromatography-mass spectrometry.

⁴ Enzyme-linked immunosorbent assay.

⁵ Gas chromatography-mass spectrometry.

Table 4

Comparison of glyphosate concentrations measured in the present and previous studies employing biological monitoring (urine sampling).

References	Type of study	Country	Analytical Method	LOD/LOQ ($\mu\text{g L}^{-1}$)	Participant numbers	Urine conc. ($\mu\text{g L}^{-1}$)	
						Statistic & Value	Max
*Current study	Occupational exposure in horticulture amenity gardening.	Ireland	LC-MS/MS ²	LOD 0.5	19 workers	AM 1.35	10.66
Mesnage et al. (2012)	Family exposure in agricultural setting	Europe	HPLC-MS ³	LOD 1 LOQ 2	1 farmer & spouse & 3 children	N/A	9.5
Curwin et al. (2007)	Farm families and 'non-farm' families	United States	Immunoassay (fluorescent microbeads)	LOD 0.9	47 fathers, 48 mothers and 117 children	AM	18
Acquavella et al. (2004)	Occupational and residential exposures in agricultural setting.	United States	HPLC following ion exchange	LOD 1	48 farmers, 48 spouses & 79 children.	1.1–2.7 GM	233
Jayasumana et al. (2015)	Investigate glyphosate levels in Sri Lankan Agricultural Nephropathy (SAN) patients	Sri Lanka	ELISA ⁴		30–3 groups of 10).	3.2 Median	≥ 80
*Rudzok et al. (2016)	Environmental exposures	Germany	validation done in comparison with GC-MS GC-MS/MS ⁵	LOQ 0.15	Group 1 Group 2 Group 3 250 samples 2–6 year old children.	56.8 73.5 3.3 Median	95th%ile
Conrad et al. (2017)	Environmental exposures	Germany	GC-MS/MS	LOQ 0.1	399 samples adults	0.14 Median	0.97
*Hoppe (2013)	European study – back-ground/dietary exposures	Europe	GC-MS/MS	LOQ 0.15	182 urine samples, 18 European countries	0.18 AM	2.80
*Markard (2014)	Environmental exposures	Germany	GC-MS/MS (presumed)	LOQ 0.15	10 male 10 female	0.21 N/A	0.65
Krüger M et al. (2014)	Human & animal to investigate environmental exposures.	Europe	ELISA partly validated against GC-MS	LOD/LOQ unknown	14 human samples	≤ 2	5
*Honeycutt and Rowlands (2014)	Environmental exposures	United States	ELISA	LOQ 7.5	35 (male, female and children).	NA	18.8

N/A: Information not available due to single measurements or not given in literature.

*Literature that has not gone through peer review or is unpublished reports.

Partial details from this table was obtained in (Niemann et al., 2015).

29% of tasks in this SEG compared to at least 50% in the other SEGs. Previous studies have associated higher pesticide exposures with the mixing and loading of pesticide concentrate (Damalas and Eleftherohorinos, 2011; Flores et al., 2011; Illing, 1997; Krenz et al., 2015; Lebailly et al., 2009; Ramos et al., 2010). In this study, the majority of the participants performed the mixing and loading of the pesticide concentrate as part of the overall task assessed (except the controlled droplet applicators who used a pre-mixed solution), therefore it was not possible to evaluate pesticide exposure during the mixing and loading of the pesticide concentrate.

A strong association was found between urinary glyphosate and fluroxypyr and the sampling time with effects being somewhat more prominent in measurements of the latter pesticide. The length of the task time for work involving the boom sprayer applicator (fluroxypyr only) ranged from 75 to 364 min, whereas pesticide application involving the manual knapsack and controlled droplet applicators groups ranged from 5 to 115 min and 33–195 min, respectively. No association between urinary pesticide levels and sampling time was found for manual knapsack applicators.

The observed stronger correlation between sampling time and urinary concentrations of fluroxypyr can likely be attributed to its lower analytical LOD, its shorter half-life (6 h compared to 33 h)

(EFSA, 2015; US EPA, 1998) and the longer duration of involved work tasks compared to glyphosate.

Study results show that there is a potential for exposure during task completed in horticulture and amenity gardening that typically uses small volumes of pesticides, ranging from 100 mL to 2 L, which warrants further investigation. *Factors that could account for the exposure reported include the reuse of PPE*; 89% of workers reused PPE such as coveralls, gloves, disposable face masks. Higher levels of urinary pesticide levels were found among workers who took breaks during the task compared those who didn't. In many cases, breaks within the task were required so that the operator could drive or walk to another field site or to take a phone call; usually PPE was not removed during such breaks. Workers who encountered problems during the pesticide task such as adjusting the nozzle, leaks or spills, change in climatic conditions or PPE issue, had higher mean levels of urinary concentration of pesticides. The sampling strategy adopted in this study i.e. spot sampling, most likely underestimates the exposure potential. Sampling times of less than 24 h, would not allow sufficient time for pesticide absorption and excretion, particularly when skin is the dominant route of exposure (Aprea et al., 2004; Flack et al., 2008; Tuomainen et al., 2002; Vitali et al., 2009). A larger study incorporating 24–72 h urine

samples would provide more reliable estimates of exposure and allow comparison with ESFA's designated ADIs and AEOLs.

6. Conclusions

Results from this study provide evidence of occupational exposure to glyphosate and fluroxypyr among amenity horticultural workers. Measured levels of urinary exposures are comparable to those reported for agricultural workers. Exposures appeared to be dependent on the sampling time and levels were higher among workers who took breaks or performed longer tasks, such as the use of either controlled droplet applicators or boom sprayers.

Further research is currently underway to investigate 24 h exposures and to evaluate dermal and inadvertent ingestion exposure and their contribution to total body burden of the pesticides.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ijheh.2017.06.008>.

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